

Three sign conventions: exact and conditional mappings

child-order sign $\longrightarrow$ local forcing sign $\longrightarrow$ event-induced flux-increment sign

Primary sign

$\epsilon_m = \sigma_m \in \{-1, +1\}$
sign of the active meta-order
in child-order event time

exact equality if

$$\begin{aligned} \mathcal{M}_{j,m} &= \sigma_m q_{j,m}, \\ q_{j,m} &\geq 0, \\ \sum_{j \in \mathcal{K}_m} q_{j,m} &> 0 \\ \epsilon_m^{\mathcal{M}} &= \epsilon_m \end{aligned}$$

Source mapping

can be ambiguous:
zero aggregate or competing local forcing

Source-based sign

$$\epsilon_m^{\mathcal{M}} = \text{sgn}(\mathcal{M}_m^*)$$

$$\mathcal{M}_m^* = \sum_{j \in \mathcal{K}_m} \mathcal{M}_{j,m}$$

aggregate forcing near the front

conditional equality if
local response is monotone,
forcing-dominated
and resolved
over the same event
 $\epsilon_m^F = \epsilon_m^{\mathcal{M}}$

Delayed or displaced response:

diffusion delay; moving-front effects; curvature

Flux-based sign

$$\epsilon_m^F = \text{sgn}(\Delta F_m)$$

$\Delta F_m = F_m^+ - F_m^-$
event-induced front-flux increment;
not the raw front flux F_m

Competing measurement effects:

background sources; discretization; event resolution

Scope control: this is a conditional conceptual map. It reports no sign-agreement percentage, no lattice/PDE result and no identification with a price or return sign.